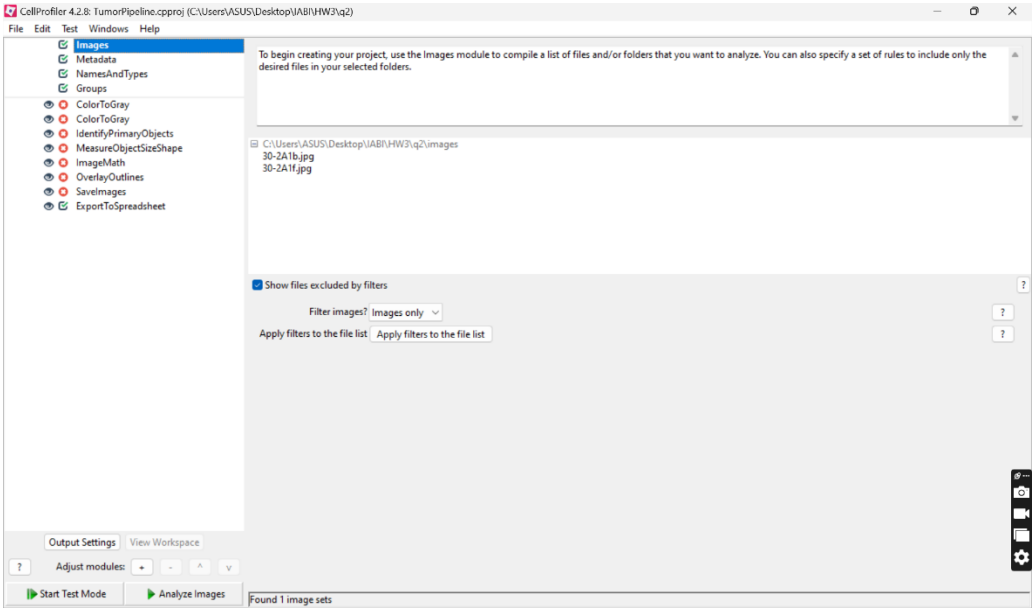
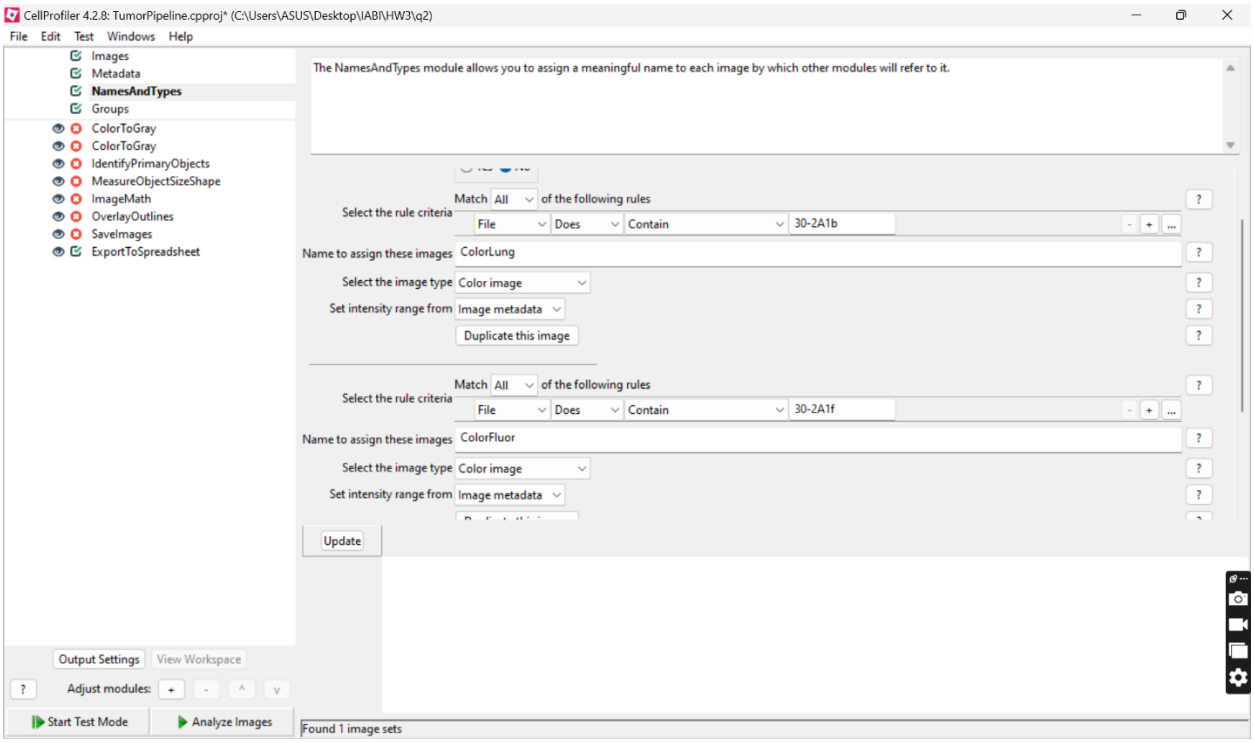
گزارش مراحل کار با CellProfiler

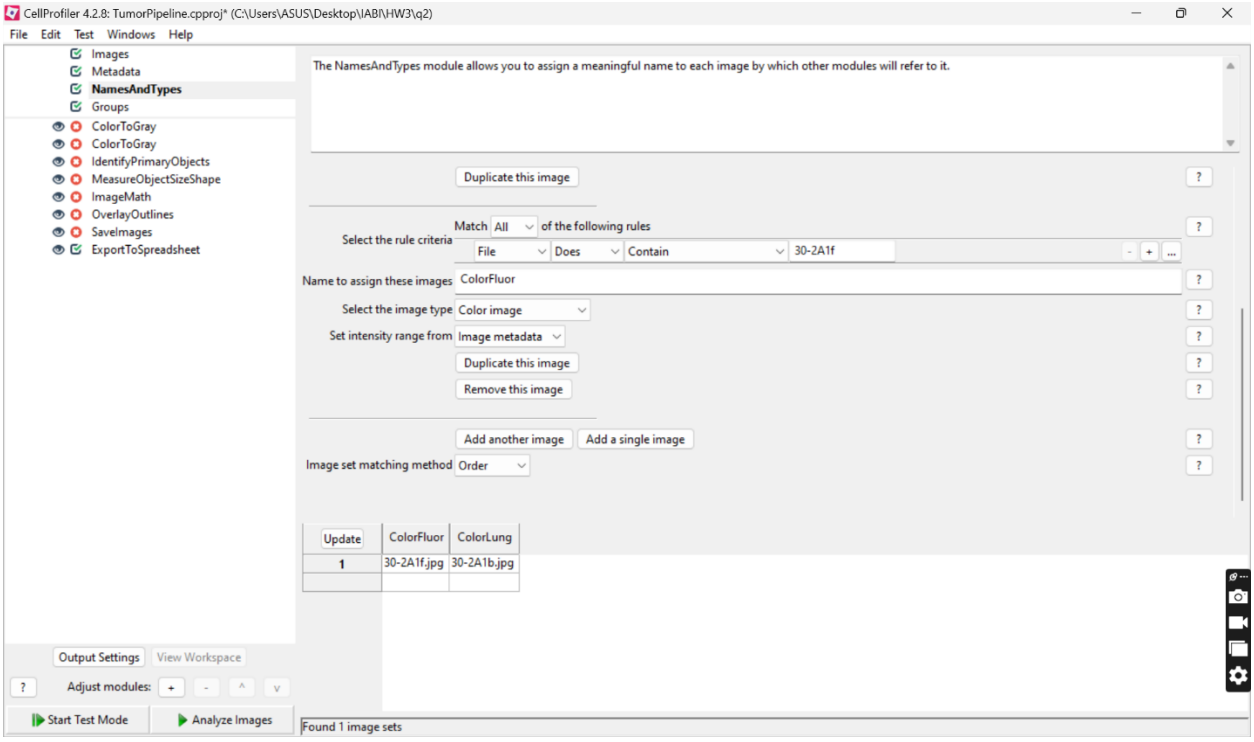
ریحانه خیاطزاده ماهانی ٤٠٢١٠٥٩٦٥

باز کردن فایل پروژه و مسیریابی صحیح به دایرکتوری شامل تصاویر:

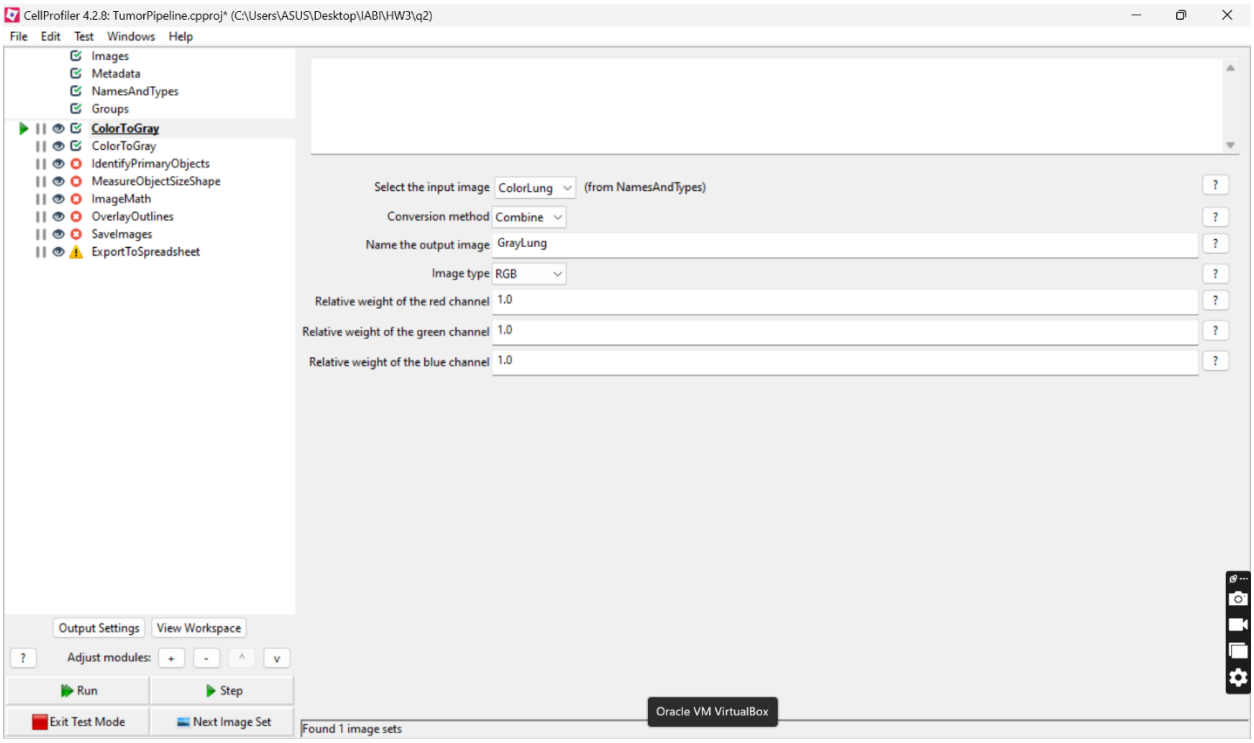


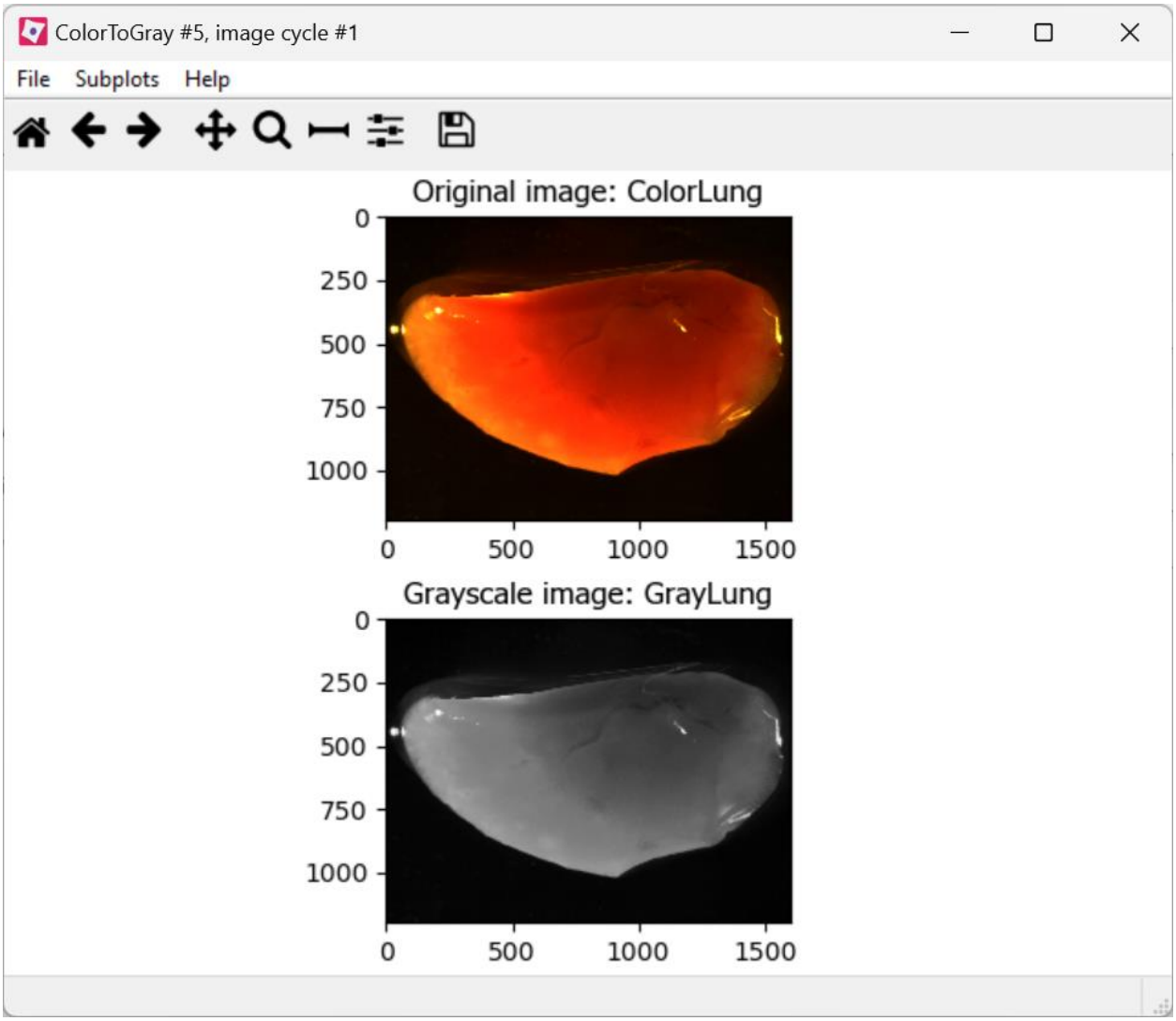
تنظیم اسم فایل تصاویر به ColorLung و ColorFluor :



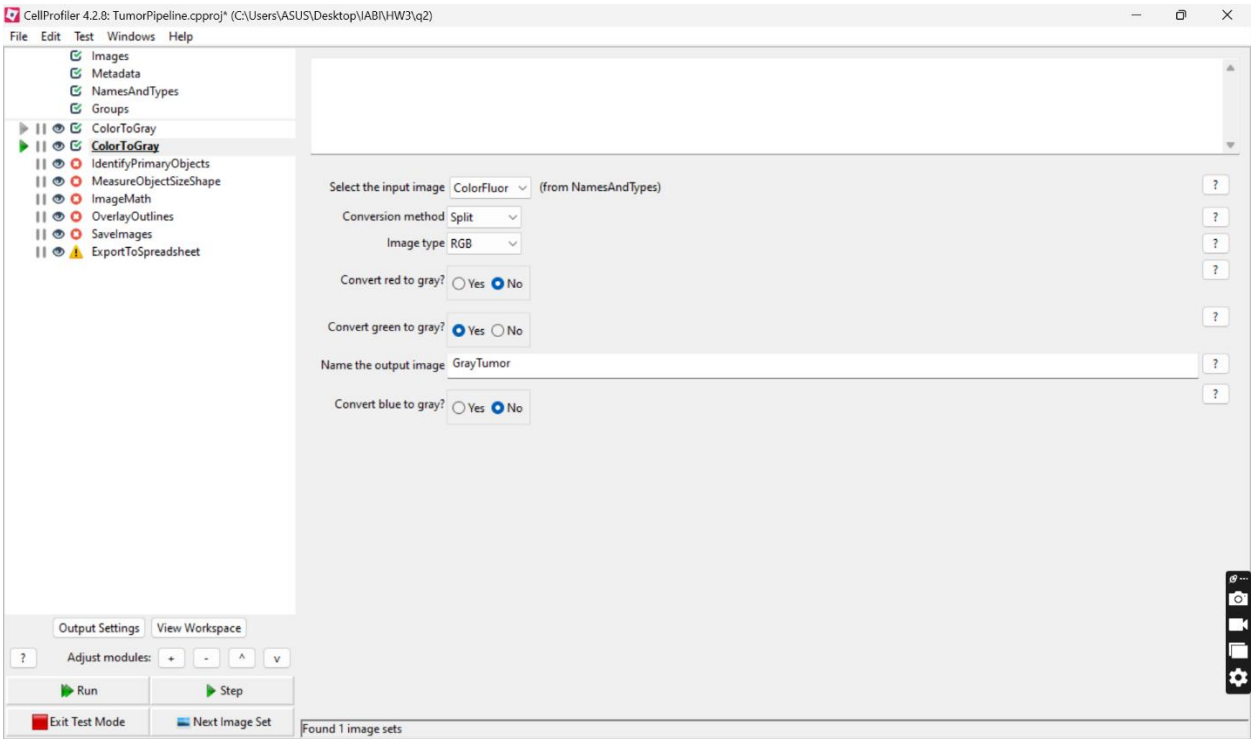


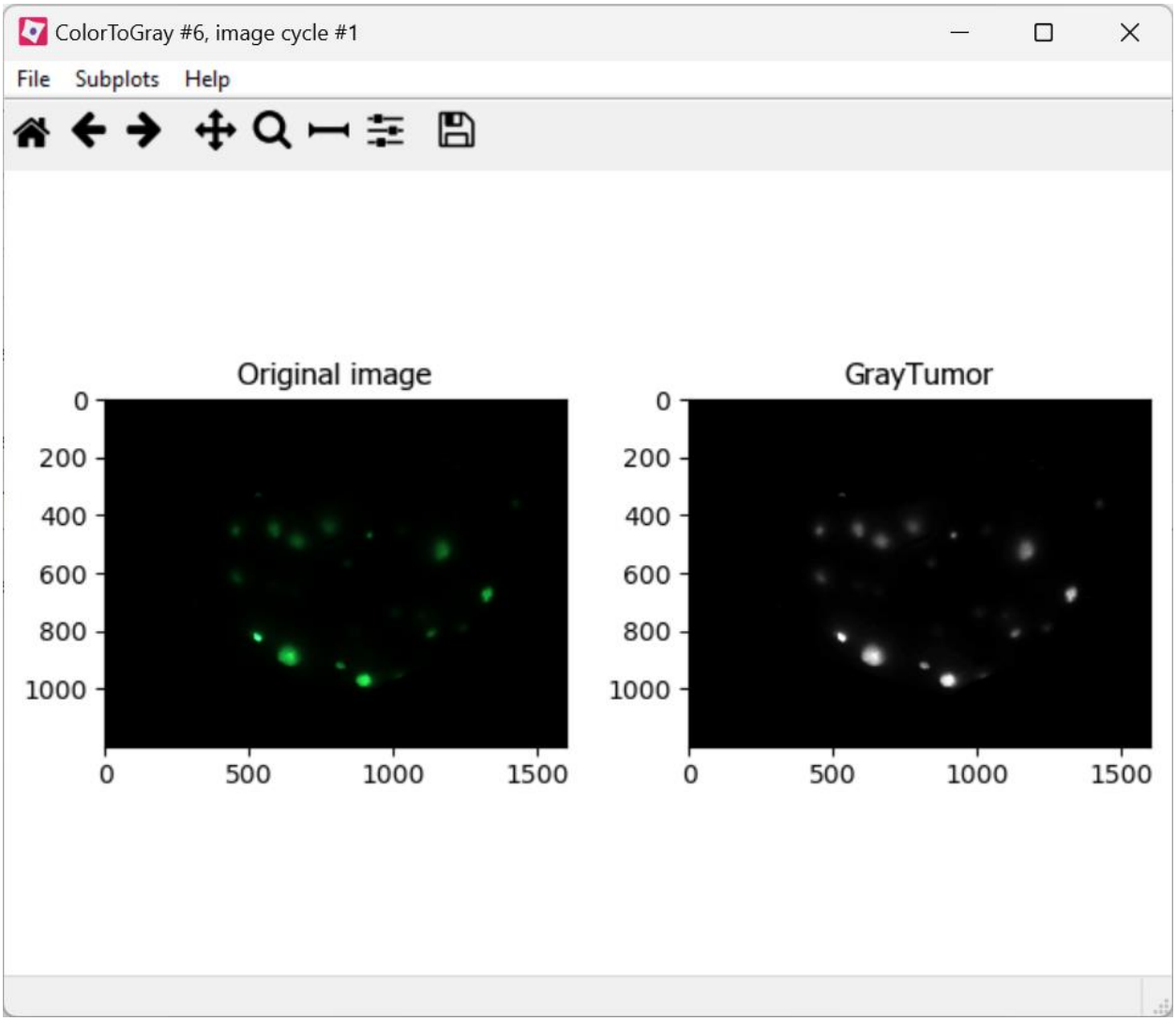
در ماژول ColorToGray اولی، تصویر ColorLung را سیاه و سفید میکنیم و با اسم GrayLung ذخیره میکنیم:


CellProfiler 4.2.8: TumorPipeline.cpproj* (C:\Users\ASUS\Desktop\IABI\HW3\q2)
File Edit Test Windows Help
Images
Metadata
NamesAndTypes
Groups
ColorToGray
ColorToGray
IdentifyPrimaryObjects
MeasureObjectSizeShape
ImageMath
OverlayOutlines
SaveImages
ExportToSpreadsheet
Select the input image ColorLung (from NamesAndTypes)
Conversion method Combine
Name the output image GrayLung
Image type RGB
Relative weight of the red channel 1.0
Relative weight of the green channel 1.0
Relative weight of the blue channel 1.0
Output Settings
View Workspace
Adjust modules:
Run
Step
Exit Test Mode
Next Image Set
Oracle VM VirtualBox
Found 1 image sets

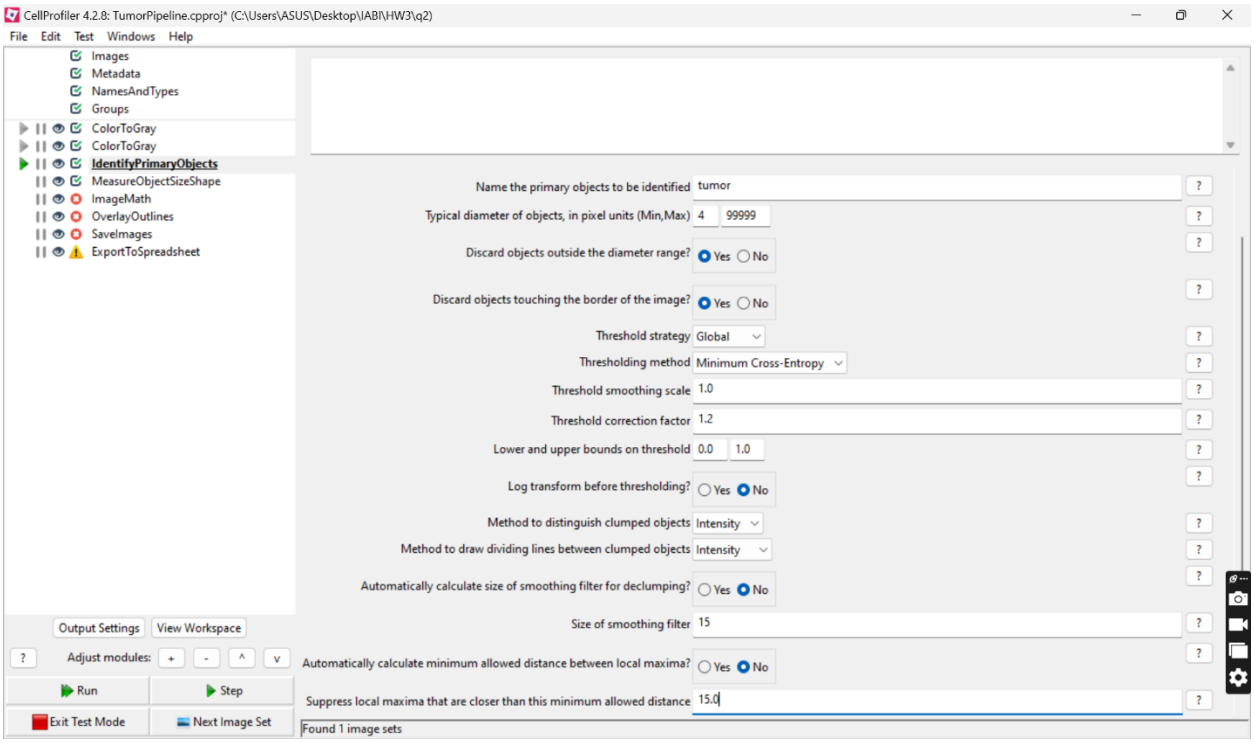


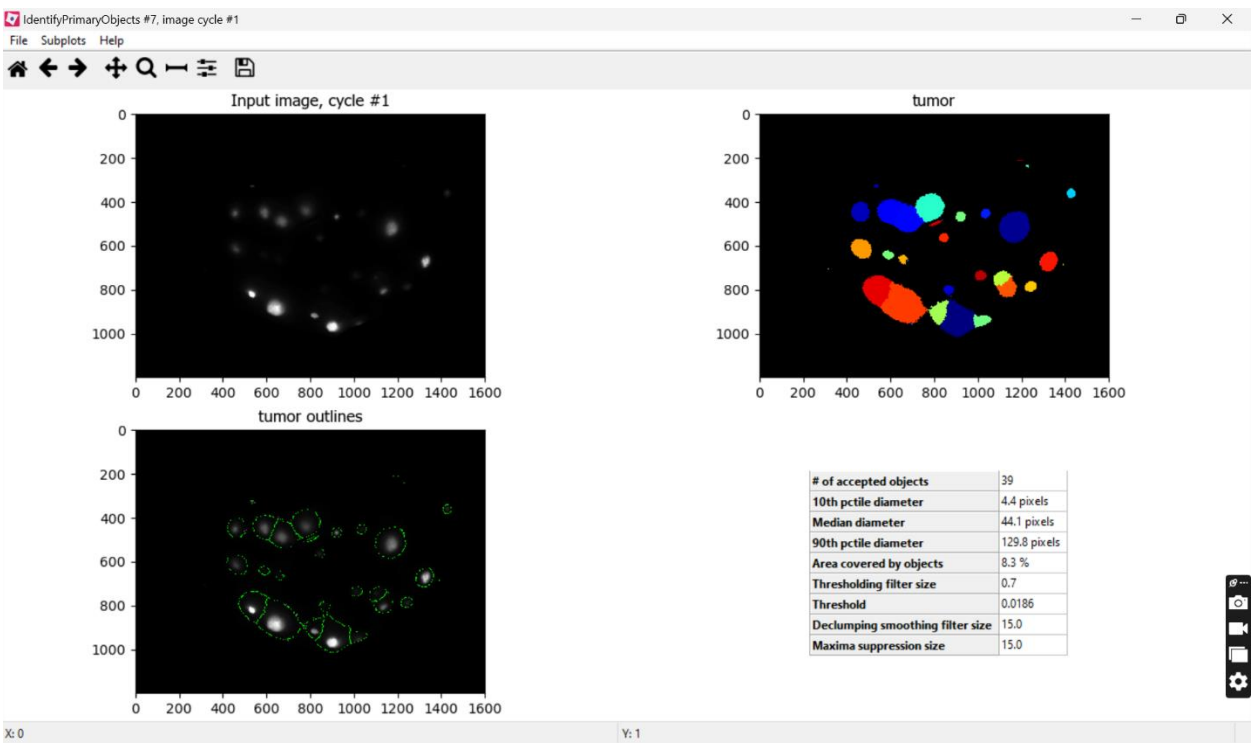
در ماژول ColorToGray دوم نیز تصویر ColorFluor را با تنظیمات زیر سیاه و سفید میکنیم و در تصویر GrayFluor ذخیره میکنیم. کانال های تصویر را به صورت split قرار میدهیم و همانطور که در تصویر اصلی دیدیم تنها کافی‌ست رنگ سبز را تبدیل کنیم. در غیر این صورت تصویر نویزی و یا کاملا سیاه خواهد شد چون نقاط توموری سبز هستند و کانال های قرمز و آبی تومور ها را نشان نخواهند داد.


CellProfiler 4.2.8: TumorPipeline.cpproj* (C:\Users\ASUS\Desktop\IABI\HW3\q2)
File Edit Test Windows Help
Images
Metadata
NamesAndTypes
Groups
ColorToGray
ColorToGray
IdentifyPrimaryObjects
MeasureObjectSizeShape
ImageMath
OverlayOutlines
SaveImages
ExportToSpreadsheet
Select the input image
ColorFluor
(from NamesAndTypes)
Conversion method
Split
Image type
RGB
Convert red to gray?
Yes
No
Convert green to gray?
Yes
No
Name the output image
GrayTumor
Convert blue to gray?
Yes
No
Output Settings
View Workspace
Adjust modules:
Run
Step
Exit Test Mode
Next Image Set
Found 1 image sets

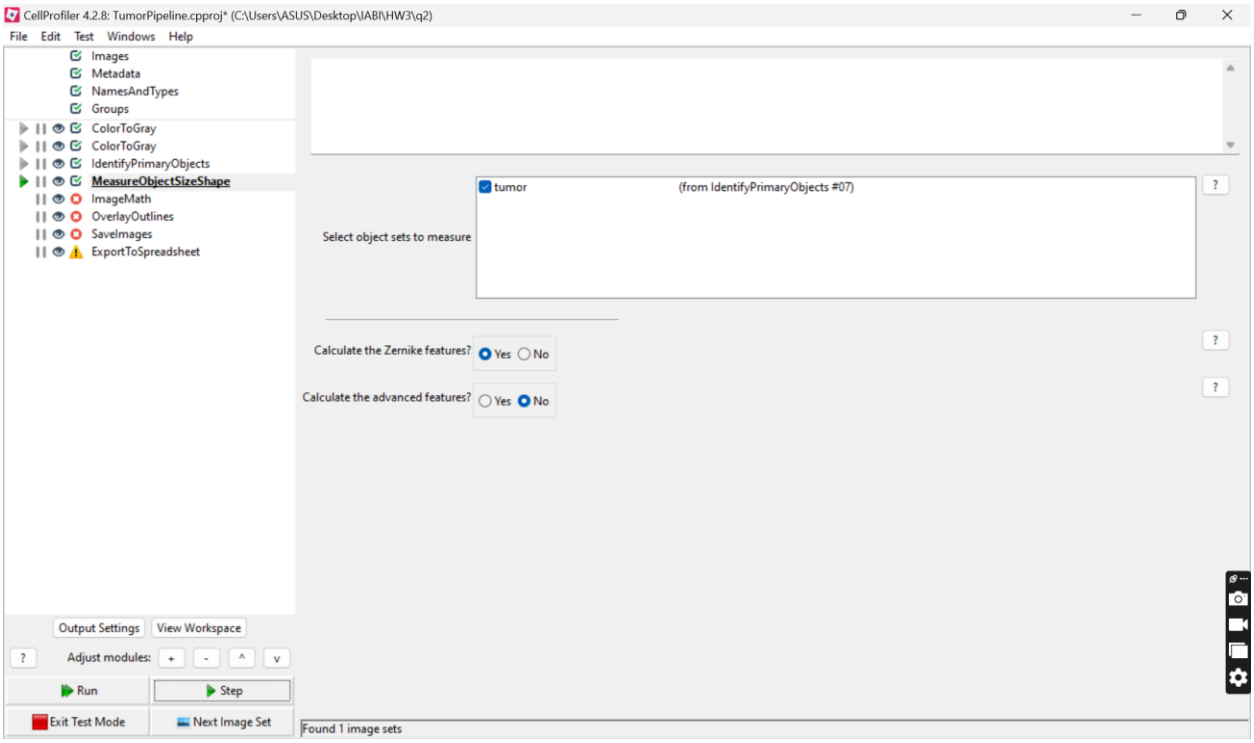


تنظیمات زیررا برای تشخیص تومور ها در ماژول IndentifyPrimaryObjects قرار میدهیم:

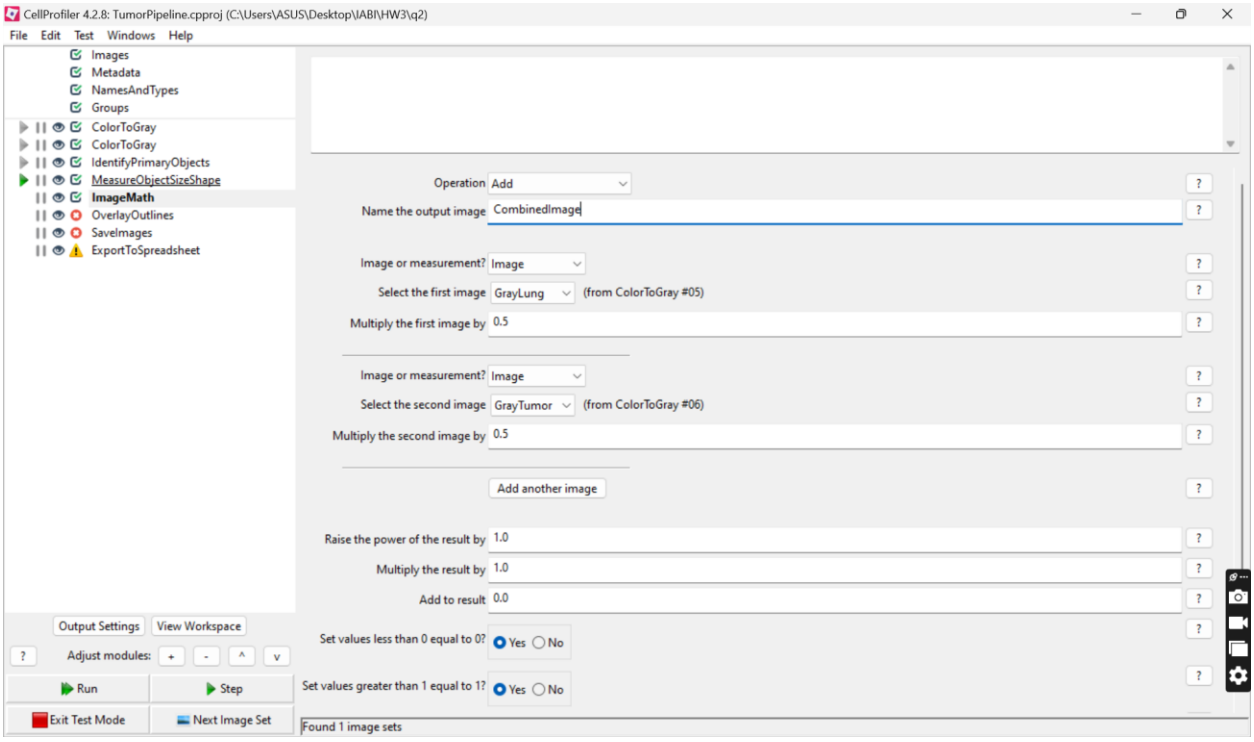




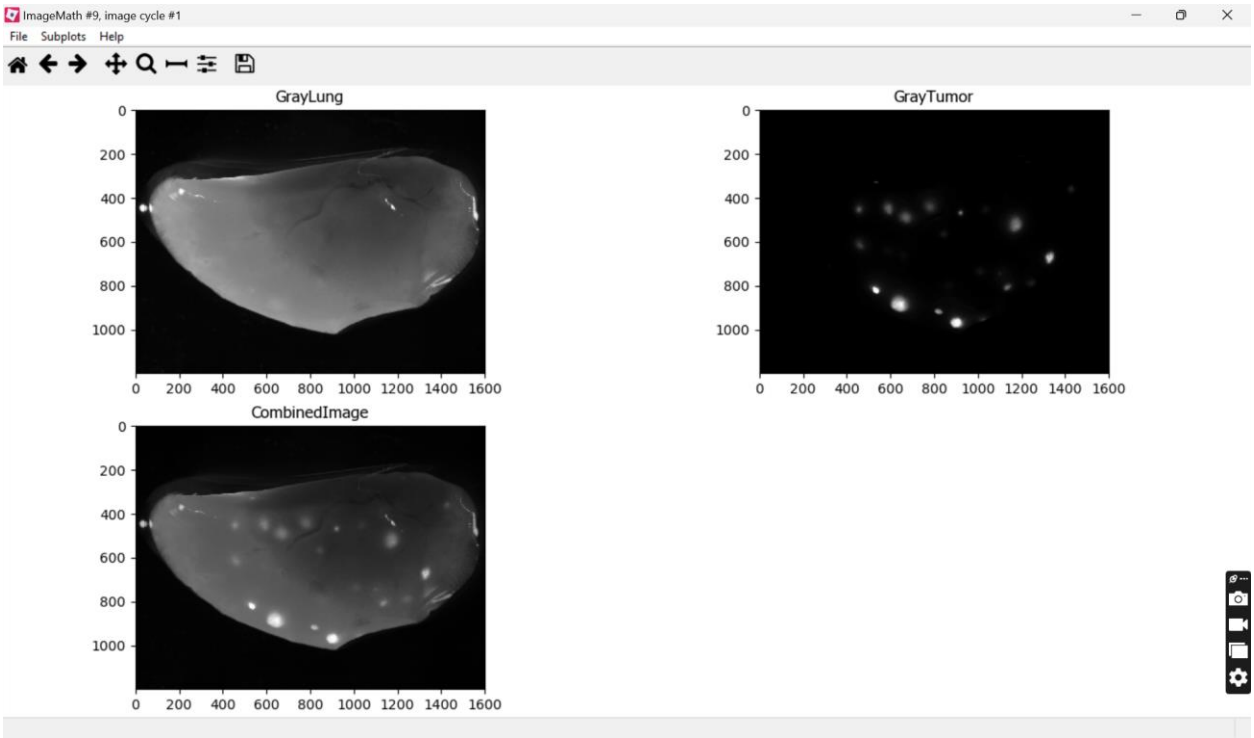
تنظیمات زیر را برای ماژول MeasureObjectSizeShape قرار میدهیم:



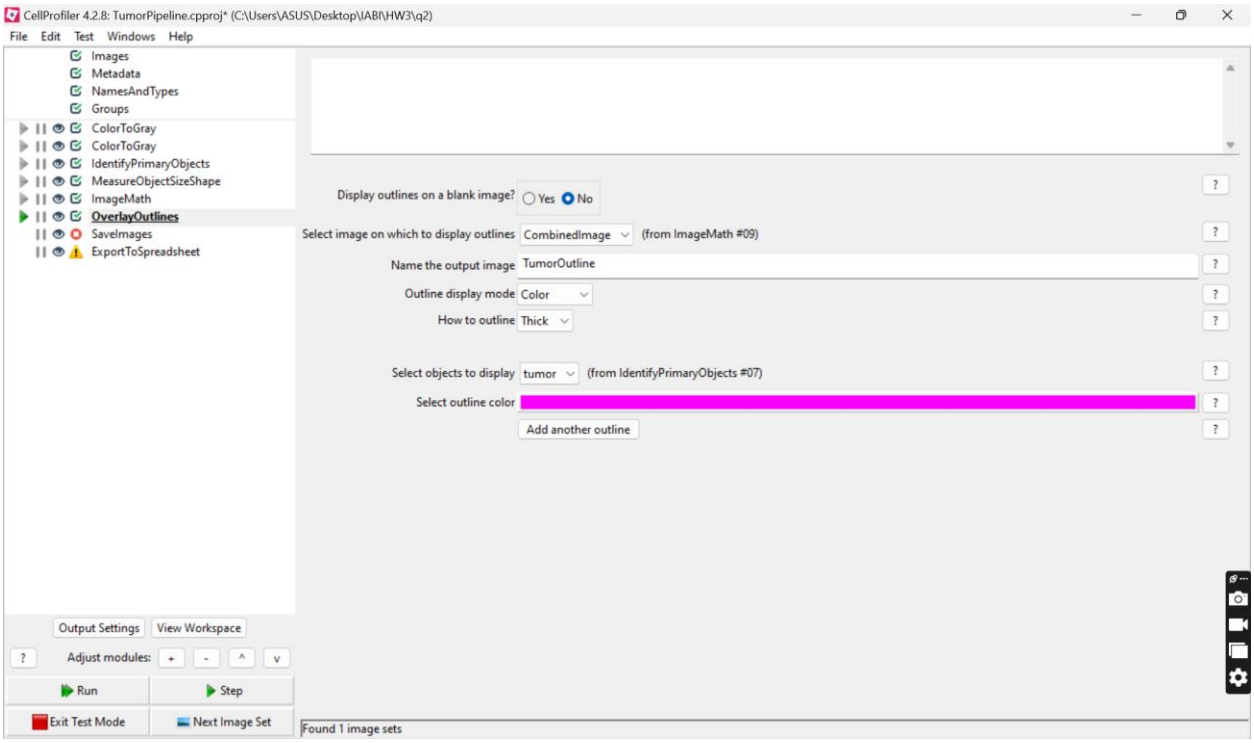
برای ترکیب تصاویر و تشخیص تومور ها در آن تنظیمات زیر را برای بخش ImageMath قرار میدهیم:

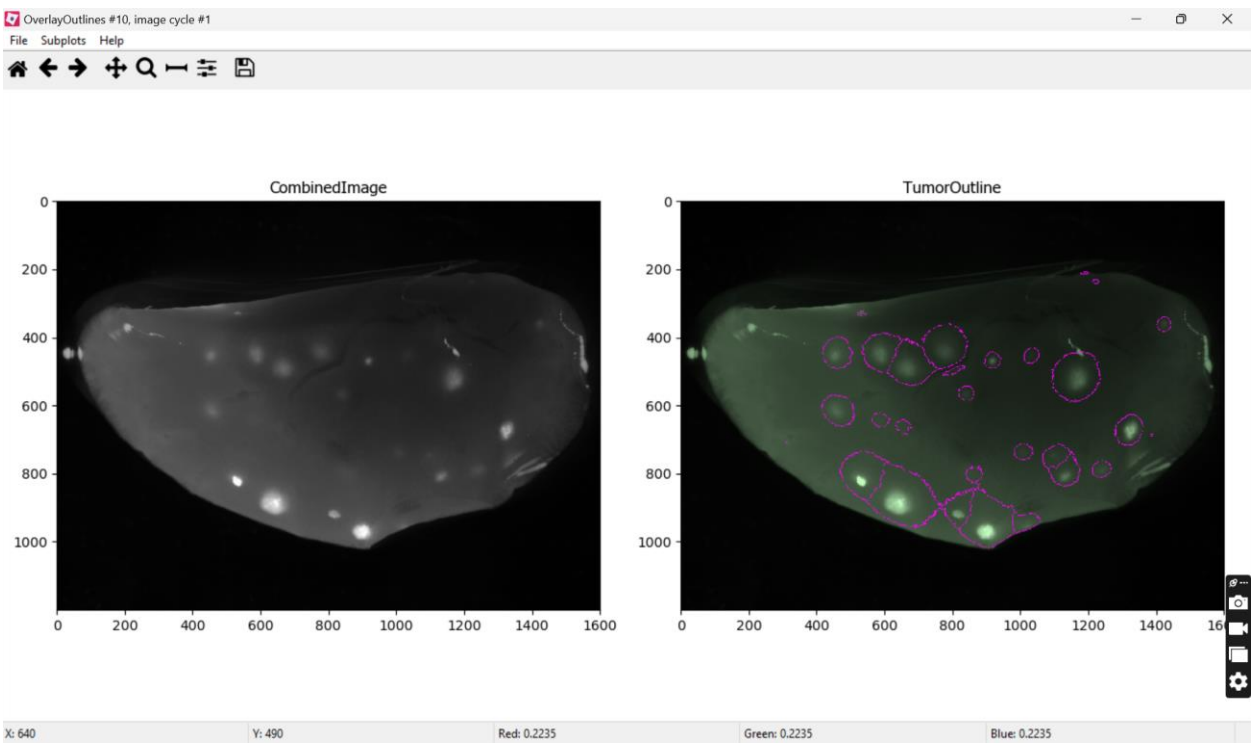


خروجی:

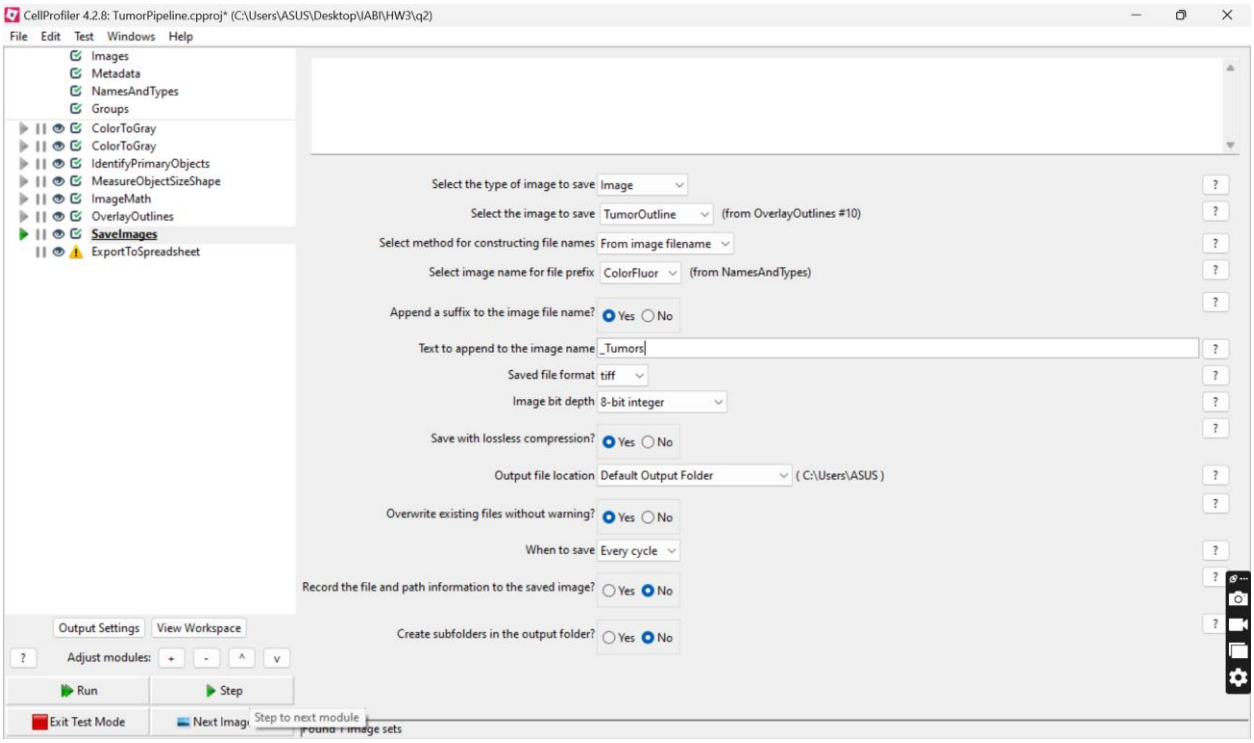


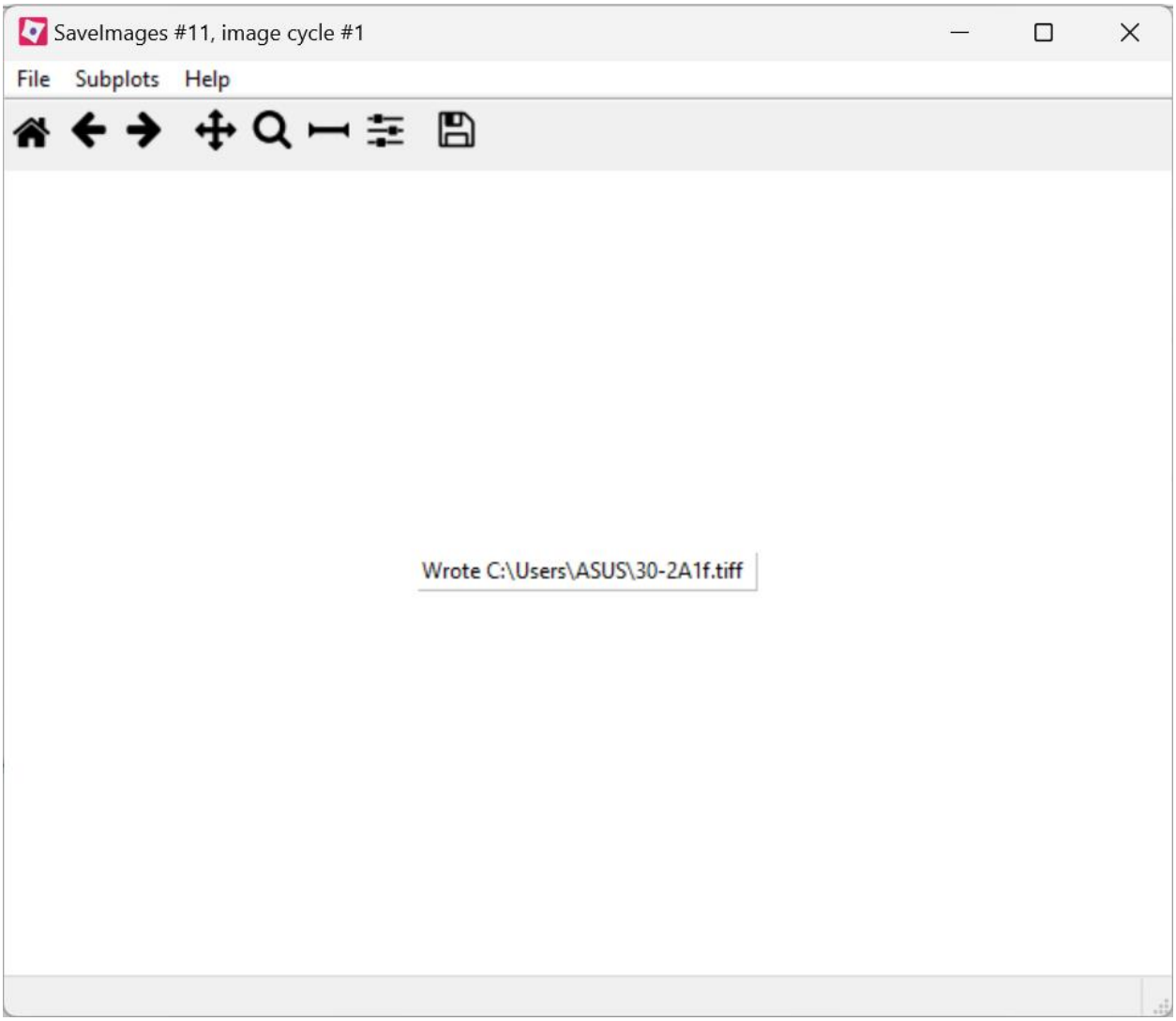
تنظیمات بخش OverlayOutline برای تشیخص تومور ها در تصویر ترکیبی:

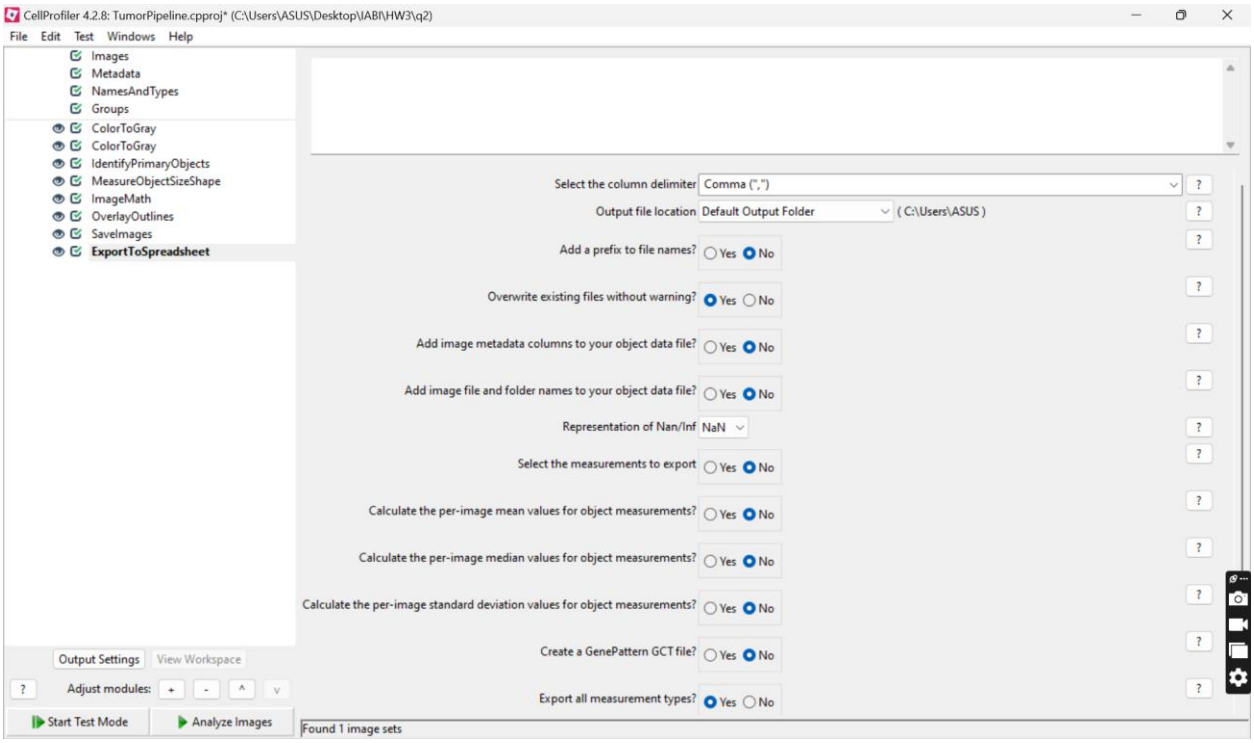

CellProfiler 4.2.8: TumorPipeline.cpproj* (C:\Users\ASUS\Desktop\IABI\HW3\q2)
File Edit Test Windows Help
Images
Metadata
NamesAndTypes
Groups
ColorToGray
ColorToGray
IdentifyPrimaryObjects
MeasureObjectSizeShape
ImageMath
OverlayOutlines
SaveImages
ExportToSpreadsheet
Display outlines on a blank image? Yes No
Select image on which to display outlines CombinedImage (from ImageMath #09)
Name the output image TumorOutline
Outline display mode Color
How to outline Thick
Select objects to display tumor (from IdentifyPrimaryObjects #07)
Select outline color
Add another outline
Output Settings
View Workspace
Adjust modules:
Run
Step
Exit Test Mode
Next Image Set
Found 1 image sets

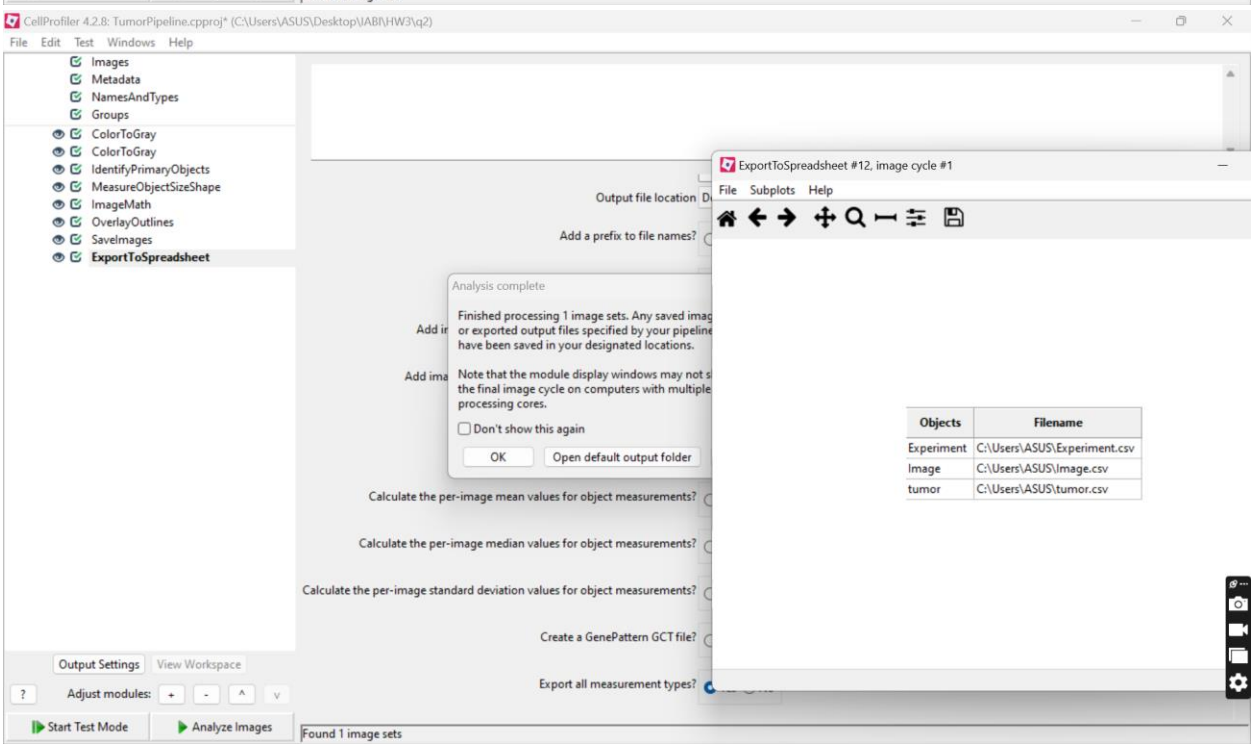

OverlayOutlines #10, image cycle #1
File Subplots Help
CombinedImage
TumorOutline
X: 640
Y: 490
Red: 0.2235
Green: 0.2235
Blue: 0.2235

تنظیمات نهایی برای خروجی گرفتن تصاویر و فایل های CSV و خروجی که در همین پوشه قرار داده شده اند.


CellProfiler 4.2.8: TumorPipeline.cpproj* (C:\Users\ASUS\Desktop\IABI\HW3\q2)
File Edit Test Windows Help
Images
Metadata
NamesAndTypes
Groups
ColorToGray
ColorToGray
IdentifyPrimaryObjects
MeasureObjectSizeShape
ImageMath
OverlayOutlines
SaveImages
ExportToSpreadsheet
Select the type of image to save Image
Select the image to save TumorOutline (from OverlayOutlines #10)
Select method for constructing file names From image filename
Select image name for file prefix ColorFluor (from NamesAndTypes)
Append a suffix to the image file name? Yes No
Text to append to the image name _Tumors
Saved file format tiff
Image bit depth 8-bit integer
Save with lossless compression? Yes No
Output file location Default Output Folder (C:\Users\ASUS)
Overwrite existing files without warning? Yes No
When to save Every cycle
Record the file and path information to the saved image? Yes No
Create subfolders in the output folder? Yes No
Output Settings
View Workspace
Adjust modules:
Run
Step
Exit Test Mode
Next Imag
Step to next module


SaveImages #11, image cycle #1
File
Subplots
Help
Wrote C:\Users\ASUS\30-2A1f.tiff


CellProfiler 4.2.8: TumorPipeline.cpproj* (C:\Users\ASUS\Desktop\IABI\HW3\q2)
File Edit Test Windows Help
Images
Metadata
NamesAndTypes
Groups
ColorToGray
ColorToGray
IdentifyPrimaryObjects
MeasureObjectSizeShape
ImageMath
OverlayOutlines
SaveImages
ExportToSpreadsheet
Select the column delimiter
Comma (",")
Output file location
Default Output Folder
(C:\Users\ASUS)
Add a prefix to file names?
Yes No
Overwrite existing files without warning?
Yes No
Add image metadata columns to your object data file?
Yes No
Add image file and folder names to your object data file?
Yes No
Representation of Nan/Inf
NaN
Select the measurements to export
Yes No
Calculate the per-image mean values for object measurements?
Yes No
Calculate the per-image median values for object measurements?
Yes No
Calculate the per-image standard deviation values for object measurements?
Yes No
Create a GenePattern GCT file?
Yes No
Export all measurement types?
Yes No
Output Settings
View Workspace
Adjust modules:
Start Test Mode
Analyze Images
Found 1 image sets


CellProfiler 4.2.8: TumorPipeline.cpproj* (C:\Users\ASUS\Desktop\IABI\HW3\q2)
File Edit Test Windows Help
Images
Metadata
NamesAndTypes
Groups
ColorToGray
ColorToGray
IdentifyPrimaryObjects
MeasureObjectSizeShape
ImageMath
OverlayOutlines
SaveImages
ExportToSpreadsheet
Analysis complete
Finished processing 1 image sets. Any saved imag
or exported output files specified by your pipeline
have been saved in your designated locations.
Note that the module display windows may not s
the final image cycle on computers with multiple
processing cores.
Don't show this again
OK
Open default output folder
ExportToSpreadsheet #12, image cycle #1
File Subplots Help
Objects
Filename
Experiment
C:\Users\ASUS\Experiment.csv
Image
C:\Users\ASUS\Image.csv
tumor
C:\Users\ASUS\tumor.csv
Output Settings
View Workspace
Adjust modules:
Start Test Mode
Analyze Images
Found 1 image sets